

Dakai Kang

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Education

University of California, Davis
Computer Science, Ph.D. Candidate

Sep 2022 – Present
Davis, CA

Research Team

Exploratory Systems Lab (expolab.org), advised by Prof. Mohammad Sadoghi (msadoghi@ucdavis.edu).

Research Topics

- High-Performance Blockchain Infrastructure
- BFT Consensus Protocols; Transaction Scheduling; Parallel Execution; MEV Problem

Teaching

- ECS165A, Databases, UC Davis, [Winter 2023](#), [2026](#) – Teaching Assistant
- ECS189F, Distributed Ledgers, UC Davis, [Fall 2023](#) – Teaching Assistant
- ECS265, Distributed Database Systems, UC Davis, [Fall 2024](#), [2025](#) – Teaching Assistant
- ECS188, Ethics in Computer Science, UC Davis, [Spring 2025](#) – Teaching Assistant

Awards

- GGCS Spring Quarter Fellowship Award, UC Davis (2023, 2024)
- GGCS Summer Ph.D. Fellowship (2023, 2024)

University of California, Davis
Computer Science, M.S.

Dec 2025
Davis, CA

University of California, Davis
GREAT (Global Research Experience in Advanced Technology) Program

Sep 2021 – July 2022
Remote

Research Assistant Intern, Exploratory Systems Lab (expolab.org).

Zhejiang University
Software Engineering, Bachelor

Sep 2018 – July 2022
Hangzhou, China

Awards

- The Third Prize Scholarship of Zhejiang University (2019, 2020, 2021)
- Excellent Engineer Scholarship of Software Engineering Major (2022)

Work Experience

Sei Labs, Inc. (<https://www.sei.io>)
Research Scientist (Incoming)

Starting in June 2026

University of California, Davis
Graduate Student Researcher, Exploratory Systems Lab (expolab.org)

Sep 2022 – Mar 2026
Davis, CA

Mysten Labs, Inc. (<https://www.mystenlabs.com>)
Research Intern

Jun 2025 – Sep 2025
Remote

- Worked on a research project on transaction scheduling in Multi-proposer consensus to accelerate parallel transaction execution and mitigate congestion under high-contention workloads.

Hangzhou Zhuxing Information Technology Co., Ltd.
Fullstack Software Engineer Intern

Jul 2020 – Oct 2020
Hangzhou, China

Publications

- [1] Dinh, T. T. A., **Kang, D.**, & Sadoghi, M. (2026, July). FEAT: Fair and Efficient Adversarial Transaction Ordering. In *Proceedings of the 2026 ACM Symposium on Principles of Distributed Computing (PODC '26)*, Egham, United Kingdom. ACM. (To appear.) <https://doi.org/10.1145/3796701.3815905>
- [2] **Kang, D.**, Chen, J., Dinh, T. T. A., & Sadoghi, M. (2025). FairDAG: consensus fairness over multi-proposer causal design. *Proceedings of the VLDB Endowment*, 19(2), 265–278.
- [3] Gupta, S., **Kang, D.**, Malkhi, D., & Sadoghi, M. (2025). *Brief announcement: Carry the tail in consensus protocols*. In *Proceedings of the 39th International Symposium on Distributed Computing (DISC 2025)* (Leibniz International Proceedings in Informatics [LIPIcs], Vol. 356, pp. 59:1–59:7). Schloss Dagstuhl – Leibniz-Zentrum für Informatik. <https://doi.org/10.4230/LIPIcs.DISC.2025.59>
- [4] **Kang, D.**, Gupta, S., Malkhi, D., & Sadoghi, M. (2025). Hotstuff-1: Linear consensus with one-phase speculation. *Proceedings of the ACM on Management of Data*, 3(3), 1–29. (**Honorable Mention for Best Artifact**)
- [5] **Kang, D.**, Rahnema, S., Hellings, J., & Sadoghi, M. (2024, May). Spotless: Concurrent rotational consensus made practical through rapid view synchronization. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)* (pp. 1916–1929). IEEE.

Preprints

- [6] Gupta, S., **Kang, D.**, Malkhi, D., & Sadoghi, M. (2025). Carry the Tail in Consensus Protocols. *arXiv preprint arXiv:2508.12173*.
- [7] Xie, S., **Kang, D.**, Lyu, H., Niu, J., & Sadoghi, M. (2025). Fides: Scalable Censorship-Resistant DAG Consensus via Trusted Components. *arXiv preprint arXiv:2501.01062*.

Open-Source Projects

- [1] ResilientDB (under Apache Incubation, <https://expolab.resilientdb.com>), **Architect**
 - Github Repository: <https://github.com/apache/incubator-resilientdb>
 - Implemented BFT consensus protocols in ResilientDB, including PBFT, HotStuff-1 and FairDAG, etc.

Peer-Review Service

Program Committee Member

- International Symposium on Reliable Distributed Systems (**SRDS**) 2026

Invited Reviewer

- IEEE International Conference on Data Engineering (**ICDE**) (Demo Track) 2026
- IEEE Transactions on Information Forensics and Security (**T-IFS**) 2025
- IEEE Transactions on Dependable and Secure Computing (**TDSC**) 2025–2026
- ACM **SIGMOD** ARI (Availability & Reproducibility Initiative) 2025
- European Conference on Computer Systems (**EuroSys**) Shadow PC program 2026

External Reviewer

- ACM Conference on Computer and Communications Security (**CCS**) 2025
- International Conference on Very Large Data Bases (**VLDB**) 2023–2025
- ACM Special Interest Group on Management of Data (**SIGMOD**) 2024–2025
- IEEE International Conference on Data Engineering (**ICDE**) 2024
- ACM/IFIP International Middleware Conference (**Middleware**) 2024